

## Adrin Alias

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### EDUCATION

University of Texas at Arlington  
Honors B.S. Mechanical Engineering

Graduation: December 2025  
3.50 GPA

### WORK EXPERIENCE

**Tesla** Palo Alto, CA  
*Hardware Test Engineering Intern* May 2025 – August 2025

- Owned the end-to-end validation for a dynamic wire harness, leading test planning, execution, and reporting across design and reliability teams to verify 2x life performance under pre-determined thermal and humidity profiles.
- Designed and fabricated an electromechanical test fixture in Catia using aluminum extrusions, sheet metal parts, linear bearings & actuators for automated mechanical testing of wire harnesses to mitigate the risk of premature field failures.
- Optimized test procedures by conducting a Design of Experiments (DOE), achieving a **90% improvement** in data accuracy for resistance measurements and streamlining post-processing efforts.
- Automated data analysis by developing a Python script to clean & graph data, cutting manual processing time by **70%**.

**Rivian** Irvine, CA  
*Manufacturing Engineering Intern* June 2023 – December 2023

- Resolved critical charge port quality issues by driving root cause analysis and implementing permanent design changes, **saving \$55,000** in tooling costs and cutting install time by **13 seconds**.
- Re-designed an aluminum fixture to align & hold air spring assemblies by analyzing part variation, modeling in Catia, creating part drawings, and validating on the line, increasing operator utilization by more than **50%**.
- Led foundational engineering analyses for new vehicle structures, including DFMA, FMEA, and tolerance stack-ups, to identify and mitigate potential design and manufacturing risks prior to production.

### PROJECTS & EXPERIENCE

**Brake Dynamometer** | Senior Design, UTA Racing, Formula SAE Arlington, TX  
*Data Acquisition & Controls Lead* August 2024 – May 2025

- Developed a custom data acquisition system (DAQ) from concept to completion for a brake dynamometer, achieving a **94% cost reduction** (\$7,700 savings) compared to commercial alternatives.
- Developed C++ firmware for a SAME51 microcontroller to process and transmit data at **20 Hz** from four sensor types (temperature, pressure, force, speed), implementing CAN bus communication for a multi-channel IR sensor.
- Designed a Python-based GUI for **real-time data visualization** and logging, enabling the characterization of brake coefficient of friction vs. temperature to validate manufacturer performance and inform design decisions.

**IoT Smart Blinds** Arlington, TX  
*Personal Project* May 2024 – July 2024

- Developed a cost-effective automated window blinds system using an ESP-32 (Arduino) and stepper motor, improving sleep quality & quantity for **under \$10** per unit.
- Prototyped a modular, compact enclosure using SolidWorks & 3D printing to house all electronics and mechanical components, ensuring adaptability across various window types.
- Programmed firmware in C++ to retrieve local sunset and sunrise times through APIs and control the motor. Implemented a database for remote control and data logging.

**Turbomachinery & Energy Systems Laboratory (TESLAB)** Arlington, TX  
*Mechanical Engineering Assistant, Part-time* March 2024 – August 2024

- Designed an 8.4 kW power distribution & control system for 4 heaters, utilizing a digital twin for optimization, custom-built sheet metal enclosures, and PID based control circuits. Cut manufacturing **costs by 30%** while maintaining performance.
- Analyzed centrifugal loading and contact pressure on a 38,000 RPM rotor using Ansys (FEA), driving design changes to eliminate stress concentrations and prevent disc slippage.
- Collaborated with a multidisciplinary team to generate turbine blades by using MATLAB to curve 2D airfoil profiles and SolidWorks to convert them into 3D models.

### SKILLS

**Software:** SolidWorks, Catia, C++, Python, MATLAB, Ansys Mechanical, MS Office, AutoCAD

**Technical:** GD&T, Rapid Prototyping, Mechanical Design, FEA, Design of Experiment (DOE), DFMA, 3D Printing